

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

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COURSE CODE: CSA0702

SLOT: B

COURSE NAME: COMPUTER NETWORKS

Question 1: Missing Layers in the Concept Map

1. Content

The given chain — Application → Presentation → ____ → Transport → Network → ____ → Physical — omits two layers of the OSI model: the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical).

Session Layer (missing 1st blank): Establishes, manages, synchronizes, and terminates the dialogue between two communicating hosts. It sets up checkpoints within a data exchange, so if a connection is interrupted, communication can resume from the last checkpoint instead of restarting entirely. This dialogue control is essential for reliability because it prevents partial or out-of-sync exchanges between applications, and it coordinates full-duplex/half-duplex communication so both ends stay synchronized.

Data Link Layer (missing 2nd blank): Takes packets from the Network layer, frames them, attaches MAC (hardware) addresses, and performs error detection (commonly CRC) before handing the resulting bits to the Physical layer. It is essential to reliability because it catches transmission errors introduced by noise on the physical medium at the very first opportunity — before those errors can propagate upward and burden higher layers (Network, Transport) with correcting problems that could have been caught locally.

Without the Session layer, applications would need to manage connection state, synchronization, and recovery themselves. Without the Data Link layer, every bit-level error on the wire would silently corrupt data delivered to the Network layer, making end-to-end reliability far more expensive to achieve. Together with the other layers, they complete an unbroken chain in which each layer's guarantees build on the one below it.

2. Image

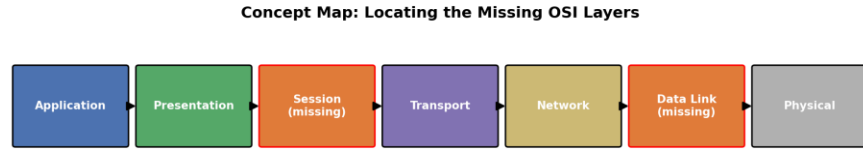


Figure Q1: Concept map with the missing Session and Data Link layers restored

3. Concept Map Architecture Table

Position	Layer	Core Function	Essential Role in Reliability
1	Application	User-facing services (HTTP, FTP, DNS)	Initiates and interprets correct requests/responses
2	Presentation	Data translation, encryption, compression	Ensures both ends interpret data identically
3 (missing)	Session	Establishes/synchronizes/terminates dialogue, checkpointing	Allows recovery from interruption without full restart
4	Transport	Segmentation, end-to-end reliability (TCP)	Guarantees ordered, error-free delivery via ACK/retransmission
5	Network	Logical addressing (IP) and routing	Ensures packets reach the correct destination network
6 (missing)	Data Link	Framing, MAC addressing, CRC error detection	Catches link-level errors before they propagate upward
7	Physical	Raw bit transmission over the medium	Provides the signal fidelity all higher layers depend on

4. Literature Insights & Evidence

- Tanenbaum & Wetherall describe the Session layer's dialogue control and checkpointing (synchronization) as the mechanism that lets long transfers recover from interruption without restarting from scratch.
- Forouzan, Data Communications and Networking, details how the Data Link layer's framing and CRC-based error detection isolate transmission noise to a single hop, which keeps errors from cascading into higher layers.
- The ISO/IEC 7498-1 reference model formally positions both layers within the seven-layer stack, confirming that skipping either breaks the intended separation of concerns.

Question 2: Data Units and Their OSI Layers

1. Content

Each OSI layer wraps data in a layer-specific Protocol Data Unit (PDU) as it moves down the stack for transmission (encapsulation), and unwraps it in reverse order as it moves up on the receiving side (decapsulation):

- Data — produced by Application/Presentation/Session layers; the raw message content before any transport-level framing.
- Segment — created at the Transport layer; adds port numbers, sequence numbers, and checksums so TCP/UDP can identify the correct application process and detect end-to-end errors.
- Packet — created at the Network layer; adds source/destination IP addresses so routers can forward it across multiple interconnected networks.
- Frame — created at the Data Link layer; adds MAC addresses and a trailer (CRC) so a switch/NIC can deliver it correctly within one physical network segment and detect corruption.
- Bits — the Physical layer's representation of the frame as electrical, optical, or radio signals suitable for the transmission medium.

This step-by-step transformation supports reliable transmission because each layer adds exactly the control information it needs to do its job — addressing, sequencing, or error-checking — without forcing every layer to solve every problem. On the receiving side, each layer strips its own header, verifies its own guarantees (e.g., Data Link checks the CRC, Transport checks sequence numbers), and passes the cleaned-up unit upward, so errors are caught as close as possible to where they occur.

2. Image

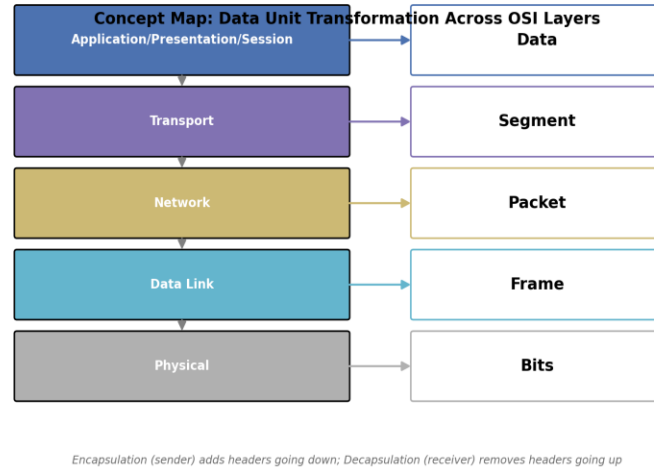


Figure Q2: Concept map of data-unit transformation from Data to Bits

3. Concept Map Architecture Table

Layer	PDU	Information Added	Purpose of the Addition
App/Presentation/Session	Data	Application-level formatting, encryption, session markers	Preserve meaning and security of the message
Transport	Segment	Source/destination port, sequence no., checksum	Identify process, order segments, detect errors end-to-end
Network	Packet	Source/destination IP address	Enable routing across multiple interconnected networks
Data Link	Frame	MAC address, CRC trailer	Local delivery on the segment and per-hop error detection
Physical	Bits	Signal encoding (voltage/light/radio)	Physically carry bits across the transmission medium

4. Literature Insights & Evidence

- Kurose & Ross describe encapsulation as the defining structural principle of layered networking, where each layer's PDU nests inside the PDU of the layer below.
- Tanenbaum & Wetherall use the same segment/packet/frame terminology to distinguish Transport, Network, and Data Link PDUs, tying each name to the header information that layer adds.
- Peterson & Davie's systems-approach treatment shows how decapsulation on the receiving side mirrors this process exactly in reverse, which is why layer-specific error checks (checksum, CRC) are performed immediately after the matching header is removed.

Question 3: MAC Address vs IP Address — Local and Global Delivery

1. Content

The Data Link layer uses MAC addresses, which are burned into network interface hardware and are only meaningful within a single physical/local network segment (a LAN). Switches use MAC addresses to deliver frames to the correct device on that segment — this is local delivery.

The Network layer uses IP addresses, which are hierarchical and globally routable. Routers use IP addresses to forward packets across multiple interconnected networks toward a distant destination — this is global (internetwork) delivery.

These two mechanisms are complementary, not redundant: IP determines which network and which final host the data must reach overall, while MAC determines the next physical hop needed to get there. At every hop along the route, the packet keeps the same source/destination IP addresses, but the MAC addresses change to reflect the current link (resolved via ARP). Only when a packet has been correctly forwarded hop-by-hop at the MAC level all the way to a router or switch that is on the same LAN as the final destination does true end-to-end delivery complete. This two-tier addressing scheme is what allows data to travel reliably from any host on any local network to any host on any other local network in the world.

2. Image

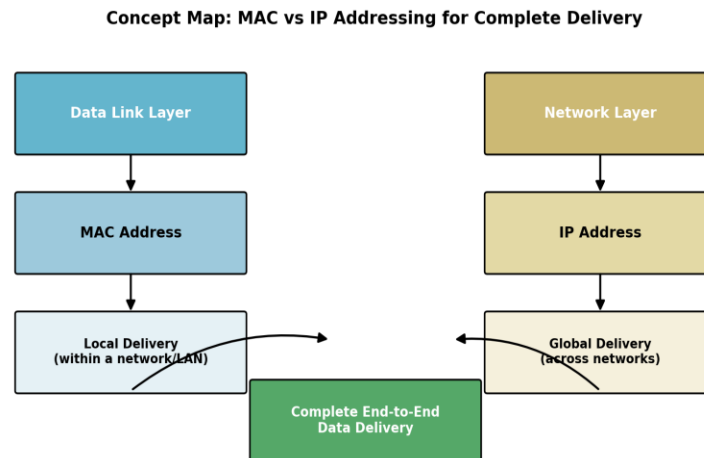


Figure Q3: Concept map linking MAC/IP addressing to local and global delivery

3. Concept Map Architecture Table

Layer	Address Type	Scope	Delivery Mechanism	Device
Data Link	MAC Address	Local (single LAN segment)	Frame switching to the correct NIC on the segment	Switch / NIC
Network	IP Address	Global (across interconnected networks)	Packet routing hop-by-hop toward the destination network	Router

4. Literature Insights & Evidence

- Forouzan explains that MAC addresses are physical and flat, valid only inside a broadcast domain, while IP addresses are logical and hierarchical, enabling scalable global routing.
- Tanenbaum & Wetherall describe the Address Resolution Protocol (ARP) as the bridge that maps a next-hop IP address to the correct local MAC address at every link along a path.
- Kurose & Ross frame this as the 'hop-by-hop link layer, end-to-end network layer' principle: IP addressing is preserved across the whole path while the link-layer address is rewritten at every hop.

Question 4: Error Detection vs Error Correction

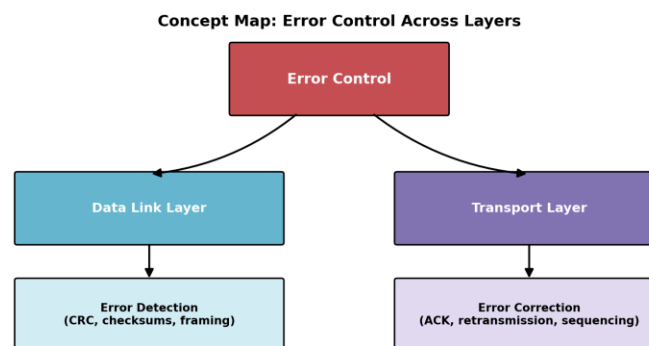
1. Content

Error control in a network is split between two layers that address different scopes of the problem:

- Data Link Layer — Error Detection: Uses CRC (cyclic redundancy check) or checksums computed over each frame to detect whether bits were corrupted while crossing a single physical link. If a frame fails this check, it is typically discarded (and, in some link protocols, a retransmission at the link level may be requested). This catches noise, interference, and hardware-level corruption immediately, at the point where it occurred.
- Transport Layer — Error Correction: Uses sequence numbers, acknowledgments (ACKs), and retransmission timers (as in TCP) to guarantee that data eventually arrives correctly and in order end-to-end, even if a segment is lost or corrupted anywhere along a multi-hop path — including failures the Data Link layer could not see, such as a router dropping a packet due to congestion.

The interaction between the two improves overall reliability because they cover complementary scopes: Data Link error detection removes the vast majority of transmission errors immediately and cheaply, hop by hop, so that only a small residual error rate ever reaches the Transport layer. Transport-layer correction then provides an end-to-end safety net that catches whatever the per-hop checks missed (or where a hop drops the data entirely), and actively recovers it through retransmission. Relying on Transport alone would be inefficient (errors would only be caught after long delay across the whole path), while relying on Data Link alone would be insufficient (it cannot recover packets lost between routers). Layering both mechanisms yields a system that is both fast to catch local errors and robust to end-to-end failures.

2. Image



Detection (Data Link) + Correction (Transport) together improve overall reliability

Figure Q4: Concept map of error detection (Data Link) vs error correction (Transport)

3. Concept Map Architecture Table

Layer	Mechanism	Scope	Technique	Outcome
Data Link	Error Detection	Single physical link (per hop)	CRC, checksum, framing	Corrupted frames discarded immediately
Transport	Error Correction	End-to-end (whole path)	Sequence numbers, ACK, retransmission timers	Guaranteed, ordered, complete delivery

4. Literature Insights & Evidence

- Tanenbaum & Wetherall distinguish error-detecting codes used at the Data Link layer from the end-to-end reliability guarantees implemented in Transport-layer protocols such as TCP.
- Kurose & Ross's treatment of 'reliable data transfer' principles shows how retransmission-based correction at the Transport layer compensates for losses that per-hop error detection cannot address, such as router-level packet drops.
- Peterson & Davie note that placing error detection close to the physical medium (Data Link) and error correction at the end hosts (Transport) reflects the end-to-end argument in system design: keep low-level mechanisms simple and fast, and place the strongest guarantees at the endpoints.

Question 5: Concept Map for Web Browsing

1. Content

Loading a webpage exercises every OSI layer in sequence: the Application layer issues an HTTP request; the Presentation layer applies TLS encryption for HTTPS and formats the data; the Session layer establishes and maintains the session with the web server; the Transport layer (TCP) breaks the request into segments and guarantees their reliable, ordered delivery; the Network layer (IP) addresses and routes those segments toward the server across the internet; the Data Link layer frames the packets for the local network (Ethernet or Wi-Fi) and resolves MAC addresses; and the Physical layer converts frames into electrical, optical, or radio signals actually sent over the medium.

Because each layer depends on the correct functioning of the layer below it, a failure at any single layer breaks the entire communication chain, regardless of whether the other layers work perfectly:

- If the Physical layer fails (e.g., cable unplugged, Wi-Fi out of range), no signal reaches the network at all — nothing above can function.
- If the Data Link layer fails (e.g., MAC/ARP resolution error), frames cannot reach the local router even though the physical medium is fine.

- If the Network layer fails (e.g., misconfigured IP, no route to the server), packets cannot reach the destination network even though the local link works.
- If the Transport layer fails (e.g., TCP handshake blocked by a firewall), no reliable connection can be established even though packets can reach the server.
- If the Application layer fails (e.g., the web server itself is down or returns an error), the browser receives no valid page even though the entire underlying network path is healthy.

This shows that web browsing reliability is only as strong as its weakest layer: OSI's layered dependency means the whole is not more reliable than its least reliable part.

2. Image

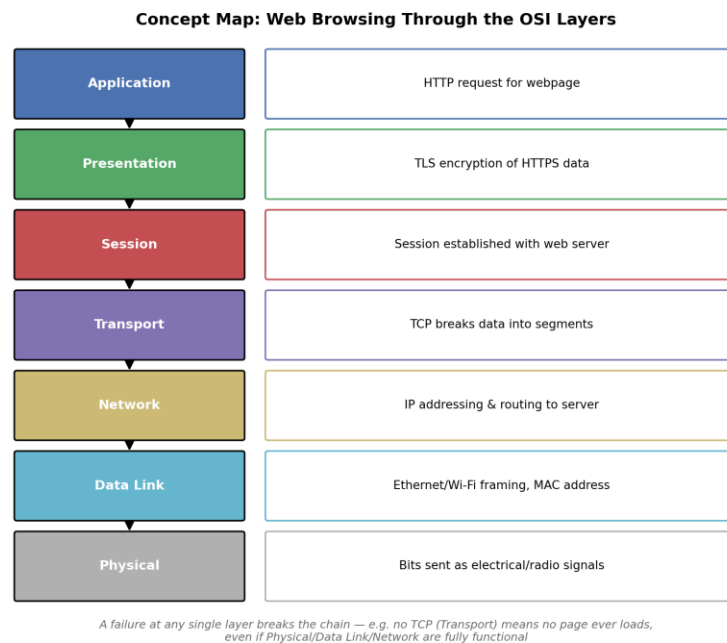


Figure Q5: Concept map of web browsing mapped across all seven OSI layers

3. Concept Map Architecture Table

Layer	Web Browsing Example	Function	Effect if This Layer Fails
Application	Browser issues HTTP GET request	Defines request/response semantics	No valid page received, even if network is healthy
Presentation	TLS encrypts HTTPS traffic	Data formatting, encryption	Data unreadable or connection rejected as insecure

Layer	Web Browsing Example	Function	Effect if This Layer Fails
Session	Session kept alive with server	Dialogue management	Session drops; repeated re-authentication needed
Transport	TCP segments the HTTP request	Reliable, ordered end-to-end delivery	Connection never establishes; page never loads
Network	IP routes packets to server	Logical addressing and routing	Packets never reach the destination network
Data Link	Ethernet/Wi-Fi frames, MAC address	Local framing and delivery	Frames never leave the local network
Physical	Signal over cable/radio	Raw bit transmission	No connectivity at all

4. Literature Insights & Evidence

- Kurose & Ross use web browsing as a canonical example to walk through every OSI/TCP-IP layer's role in a single HTTP request-response cycle.
- Tanenbaum & Wetherall's layered-service model formalizes the principle that each layer only uses services from the layer directly below it, which is precisely why a failure at any one layer cannot be bypassed by the layers above.
- Peterson & Davie's systems-approach discussion of protocol layering describes this dependency chain as the basis for both the model's modularity and its single point-of-failure characteristic at each layer boundary.

Question 6: Troubleshooting Using the Concept Map

1. Content

The concept map shows: Physical check → Data Link check → Network fail → Transport fail → Application fail. Because the OSI layers are strictly dependent — each layer relies on the one below it — this pattern indicates that the fault originates at the Network layer, and the failures shown at Transport and Application are consequences of that single root cause, not independent problems.

Since Physical and Data Link are both functioning, the physical medium is intact and local frame delivery within the LAN is working (cabling, NIC, switch, and MAC-level communication are healthy). The break occurs exactly at the Network layer, which points to causes such as: a misconfigured IP address or subnet mask, a missing or incorrect default gateway, a broken routing table entry, an unreachable next-hop router, or an access-control/firewall rule blocking IP traffic. Because Transport (TCP/UDP) and Application depend entirely on Network-layer delivery to function, they fail automatically once Network-layer delivery is broken — there is no need to separately diagnose them.

This is exactly why a concept map is valuable for troubleshooting: it lets an engineer isolate the fault boundary quickly by checking layers from the bottom up. Since everything below Network

is confirmed working, diagnostic effort should focus specifically on Network-layer tools and configuration — such as checking IP configuration, testing with ping/traceroute to the gateway, and inspecting routing tables — rather than wastefully re-checking cabling or application settings that are not the actual source of the problem.

2. Image

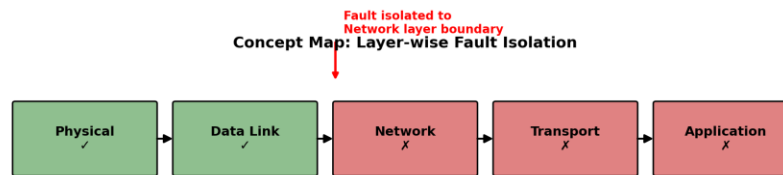


Figure Q6: Concept map isolating the fault at the Network layer boundary

3. Concept Map Architecture Table

Layer	Status	Likely Cause if Faulty	Diagnostic Tool	Troubleshooting Action
Physical	OK	N/A	Cable/link light check	Confirmed working — no action needed
Data Link	OK	N/A	ARP table, switch port status	Confirmed working — no action needed
Network	Fault	Wrong IP/subnet, missing gateway, bad route, firewall block	ping, traceroute, ipconfig/ifconfig, routing table	Verify IP config, gateway reachability, and routing
Transport	Fault (consequence)	No connection possible without Network delivery	netstat, port scan (after fixing Network)	Re-test only after Network layer is restored
Application	Fault (consequence)	No data can reach the application without lower layers	Application logs (after fixing Network)	Re-test only after Network layer is restored

4. Literature Insights & Evidence

- Tanenbaum & Wetherall's layered architecture principle — that each layer depends strictly on the services of the layer beneath it — is the theoretical basis for bottom-up fault isolation used in real network troubleshooting.

- Forouzan's networking texts describe standard bottom-up troubleshooting methodology (physical cabling and link lights checked first, then addressing/routing, then transport/application), matching the diagnostic order implied by this concept map.
- Cisco networking-academy materials and ITU-T's OSI-based fault-management guidance similarly recommend isolating the lowest failing layer first, since higher-layer symptoms are typically downstream effects rather than separate faults.